back in, after it was banished by the Newtonian clockwork paradigm of classical mechanics?

There's so much wrong with this that it's hard to know where to start. First, "free will" versus "determinism" isn't the right distinction to draw. Determinism should be opposed to "indeterminism," and free will should be opposed to "no free will." Determinism is straightforward to define: given the exact current state of the system, the laws of physics determine precisely the state at later times. Free will is trickier. One usually hears free will defined as something like "the ability to have chosen otherwise." That means we're comparing what really happened (we were in a situation, we made a decision, and we acted accordingly) to a different hypothetical scenario (we wind the clock backward to the original situation, and ask whether we "could have" decided differently). When playing this game, it's crucial to specify exactly what is kept fixed between the real and hypothetical situations. Is it absolutely everything, down to the last microscopic detail? Or do we just imagine fixing our available macroscopic information, allowing for variation within invisible microscopic details?

Let's say we're hard-core about this question, and compare what actually happened to a hypothetical re-running of the universe starting from exactly the same initial condition, down to the precise state of every last elementary particle. In a classical deterministic universe the outcome would be precisely the same, so there's no possibility you could have "made a different decision." By contrast, according to textbook quantum mechanics, an element of randomness is introduced, so we can't confidently predict exactly the same future outcome from the same initial conditions.

But that has nothing to do with free will. A different outcome doesn't mean we manifested some kind of personal, supra-physical volitional influence over the laws of nature. It just means that some unpredictable quantum random numbers came up differently. What matters for the traditional "strong" notion of free will is not whether we are subject to deterministic laws of nature, but whether we are subject to impersonal laws of any sort. The fact that we can't predict the future isn't the same as the idea that we are free to bring it about. Even in textbook quantum mechanics, human beings are still collections of particles and fields obeying the laws of physics.

For that matter, quantum mechanics is not necessarily indeterministic. Many-Worlds is a counterexample. You evolve, perfectly deterministically,